

L-1. Rehabilitation of Degraded Lands

 Degraded Soil & Land Improvement	 High	 High		 Term
Measure Type	Cost	Preferences by Community	Hazards	Implementation Time of Measures

Descriptions

Measures	Rehabilitation of degraded land refers to the process of restoring land that has been damaged or degraded by human activities or natural processes, to a healthier, more productive, or sustainable state. This can entail a range of methods and approaches meant to enhance the biological, chemical, and physical characteristics of the land. Deforestation, excessive grazing, bad farming methods, mining, pollution, and climate change are some of the causes of land degradation.								
	Land degradation is a major problem in Bangladesh's CHT, mostly as a result of deforestation, shifting cultivation (referred to locally as "Jhum cultivation"), soil erosion, and population pressures. The area has issues with sustainable environmental practices and land usage because of its rugged topography, delicate ecosystem, and varied indigenous populations. Reversing the detrimental effects on the environment and advancing sustainable development in this situation depend heavily on effective rehabilitation methods. This strategy needs to be context-specific and community-driven in order to take into account the region's distinct cultural and environmental characteristics.								
Location	R-114	R-99	R-65	K-98	K-76	K-71	B-173	B-77	B-76
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Applicable Sites	- Deforested Hillslopes – Areas degraded by slash-and-burn (Jhum) cultivation, logging, and encroachment. - Eroded Riverbanks and Valley Floors – Sites affected by sedimentation, soil loss, and flooding. - Abandoned Agricultural Lands – Depleted lands requiring soil restoration and vegetation cover.								
Slope	Less than 3%		3% to 15%		15% to 30%		30% to 60%		Above 60%
	√		√		√		√		
Design options and performance	- Combines tree planting with crop cultivation, enhancing soil fertility and erosion control. - Reduces runoff and soil loss, improving water retention for crops. - Promotes native forest regrowth, improving biodiversity and climate resilience. - Includes mulching, green manure application, and check dams to stabilize degraded land.								
Feasibility criteria	Technical Feasibility Ensures effective soil conservation and reforestation success.								

	- Determines appropriate rehabilitation methods. Environmental Feasibility - Prevents further degradation and enhances land productivity. - Supports habitat restoration for native species. - Mitigates climate change impacts through afforestation. Economic Feasibility - Utilizes locally available materials and labor. - Enhances food security and livelihoods. - Encourages self-sustaining land management techniques. Social Feasibility - Engages local populations in construction and management. - Supports farming, fisheries, and household water supply.			
Operation and maintenance	Routine Land Management – Regular weeding, erosion control, and vegetation monitoring. Soil Fertility Enhancement – Application of organic compost, green manure, and soil conditioners. Seasonal Adjustments – Rainwater harvesting during monsoons, controlled grazing during dry periods. Community-Based Monitoring – Establishing local forest management groups for sustained oversight.			
Cost	High			
Benefits				
	Type	Low	Medium	High
	Canopy Cover Increase			
	Carbon Sequestration			
	Crop Production Increase			
	Crop Failure Risk Reduction			
	Ensuring Food Security			
	Fruit Production Increase			
	Forestry Goods Production Increase			
	Livestock Production Increase			
	Fisheries Production Increase			
	Alternative Livelihood Generation			
	Increase of Income			
	Soil Erosion Control			
	Improved Soil Health			
	Soil Fertility Increase			
	Sustainable Water Supply			
	Water Quality Improvement			
	Water Purification			
	River Bank Erosion Control			
	Surface Runoff Reduction			

	Groundwater Recharge			
	Pest and Disease Management			
	Reduction of Pesticides and Fertilizer Use			
	Ecosystem and Biodiversity Increase			
	Better Communications			
	Better Education			
	Better WASH facilities			
Implementation Challenges				
	Type	Low	Medium	High
	Land unavailable/high land area required			
	Supply chain disconnection			
	Land right related problems			
	High human investment capital need			
	Insufficient infrastructures			
	Less demand of agroforestry product			
	Environmentally unsustainable			
	Socially unacceptable			
	Less beneficial			
Environmental performance	• Soil Erosion Control – Prevents landslides and loss of fertile topsoil. • Carbon Sequestration & Climate Resilience – Reduces carbon footprint through afforestation. • Water Retention Improvement – Enhances groundwater recharge and reduces runoff. • Biodiversity Conservation – Restores native vegetation and wildlife habitats.			
Remarks	Rehabilitation methods must be tailored to specific land conditions and community needs to ensure effectiveness and long-term success. Community-led restoration plays a crucial role in sustainability, as engaging local people fosters a sense of ownership and commitment to maintaining restored landscapes. Integrating traditional knowledge with scientific approaches enhances resilience, combining indigenous land management practices with modern techniques for greater impact.			